

Taresh Guleria

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RESEARCH FOCUS

Additive manufacturing of functional metallic materials — laser powder bed fusion of NiTi and NiTiHf shape memory alloys, functionally graded architectures, process–structure–property relationships, and in-situ process monitoring for qualification. Seven peer-reviewed publications including *Acta Materialia*, *Materials Today*, and *Additive Manufacturing*, with research funded by NSF, DoD, AFRL, America Makes, and conducted in collaboration with Oak Ridge, Los Alamos, and Sandia National Laboratories.

EDUCATION

Ph.D., Industrial Engineering

Expected Dec 2026

Texas A&M University — GPA 4.0/4.0

College Station, TX

M.S., Industrial and Systems Engineering

Dec 2024

Texas A&M University — GPA 3.45/4.0

College Station, TX

M.Engg., Production Engineering

June 2020

Thapar Institute of Engineering and Technology — GPA 8.56/10 — Top 5% of class

Punjab, India

B.Tech., Mechanical Engineering

June 2018

SRM Institute of Science and Technology — GPA 7.29/10

Chennai, India

RESEARCH EXPERIENCE

Graduate Research Assistant, Ph.D.

Jan 2023 – Present

Texas A&M University — Dept. of Industrial & Systems Engineering

College Station, TX

- Lead NiTi and NiTiHf alloy development on a 3D Systems ProX 200 LPBF platform, establishing optimized process windows for defect-free builds.
- Built data-driven printability maps enabling defect-free processing of functionally graded NiTi-based alloys under NSF-funded research.
- Directed a Los Alamos National Laboratory project on NiTi lattice structures and sealing-ring development for radioactive material containment.
- Designed and validated active NiTi interlocking metasurfaces with Sandia National Laboratories using FEA and DIC, demonstrating repeatable shape recovery, high locking force, and cyclic stability.
- Quantified the effect of vector scan length and part geometry on melt pool thermal behavior via in-situ process monitoring, establishing scan strategy guidelines adopted across lab builds.

R&D Intern — Metals R&D & Government Relations

Jan 2026 – Aug 2026

EOS of North America

Pflugerville, TX

- Led R&D experimental design on superelastic NiTi components, driving development from experimental design through production-ready builds and accelerating time-to-qualification for aerospace and medical customer programs.
- Designed LPBF builds for the DELTA-QUAL and ACCELERATE (America Makes) programs that cut re-qualification coupons by ~80% while meeting mechanical and microstructural targets, using an in-situ monitoring-integrated build strategy portable across OEM platforms.
- Delivered closed-loop powder reuse methodology on the RECAP (America Makes) program, reducing virgin powder consumption and waste by ~75% and improving cost and sustainability at production scale.
- Initiated the VITAL project (LIFT): a closed-loop LPBF control and property-prediction framework aligning qualification strategies across multiple OEM machine platforms.
- Co-authored project proposals to DoD, AFRL, and America Makes supporting a \$2M+ portfolio of funded and submitted work.

Research Intern (NSF INTERN Fellow) — Manufacturing Demonstration Facility

May 2024 – Aug 2024

Oak Ridge National Laboratory

Knoxville, TN

- Led a shape memory alloy project on LPBF pseudoelastic NiTi, characterizing anisotropic tensile and cyclic behavior driven by grain morphology and crystallographic texture.
- Performed EBSD and in-situ XRD linking process-controlled microstructure and martensitic transformation to strain localization and pseudoelastic response.
- Achieved >99% dense NiTi by tuning LPBF parameters and mapped process–structure–property relations for deformation banding and cyclic stability under multi-cycle loading.
- Delivered microstructure-control strategies that informed MDF's roadmap for scaling NiTi LPBF toward manufacturing scale.

Graduate Engineer Trainee — Oil and Gas Sector

Aug 2017 – Mar 2018

Siemens

Gurugram, India

- Implemented BIM workflows for a gas-based power plant project in Berlin and designed ductile piping systems using AVEVA-PDMS and CREO.
- Ran computational evaluations of oil and gas pipe wall thickness for stress resistance under operational loads.

PUBLICATIONS (7 PEER-REVIEWED · 2 FIRST-AUTHOR · 2 UNDER REVIEW OR IN PREPARATION)

1. **Guleria, T.**, et al. "Beyond the material-per-function paradigm: functionally graded active structures via LPBF of NiTi." *Additive Manufacturing*, 2026. doi:10.1016/j.addma.2026.105276
2. **Guleria, T.**, et al. "Adaptive sealing structures with functionally graded NiTi SMA rings." *Materials & Design*, under review, 2026.
3. **Guleria, T.**, et al. "Effect of scan length and geometry on thermal history of NiTi melt pools in LPBF." In preparation.
4. Elsayed, A., **Guleria, T.**, et al. "Functionally graded NiTiHf high-temperature SMAs via LPBF." *Acta Materialia*, 2025. doi:10.1016/j.actamat.2025.121175
5. Elsayed, A., **Guleria, T.**, et al. "Superelastic interlocking metasurfaces enabled by additive manufacturing." *Materials Today*, 2026. doi:10.1016/j.mattod.2026.103369
6. Elsayed, A., **Guleria, T.**, et al. "Active interlocking metasurfaces enabled by shape memory alloys." *Materials & Design*, 244, 2024. doi:10.1016/j.matdes.2024.113137
7. Fernandez-Zelaia, P., ..., **Guleria, T.**, et al. "Microstructure-sensitive mechanical behavior of additively manufactured pseudoelastic SMA." *Additive Manufacturing Letters*, 2025. doi:10.1016/j.addlet.2025.100270
8. **Guleria, T.**, Verma, Zafar, Jain. "Kevlar-reinforced UHMWPE composite for body armor." *J. Reinforced Plastics and Composites*, 2021. doi:10.1177/0731684420959449
9. Lawrence, ..., **Guleria, T.**, et al. "Cybersecurity and data science curriculum for secondary student computing programs." doi:10.22369/issn.2153-4136/14/2/2

CONFERENCE PRESENTATIONS

- ASTM International Conference on Advanced Manufacturing, Las Vegas, NV, 2025 — lead author, AM of functional NiTi alloys
- Solid Freeform Fabrication Symposium, Austin, TX, 2025 — co-author, mechanical behavior of AM pseudoelastic Nitinol
- International Conference on Composites Manufacturing, IIT Mandi, 2022

TECHNICAL EXPERTISE

AM Systems: LPBF (EOS M290, M400-1/4, AMCM M290-2, 3D Systems ProX 200), in-situ process monitoring, IR cameras, optical tomography, adaptive/closed-loop process control

Process Development: Printability mapping, parameter optimization, thermal history analysis, defect mitigation, production qualification strategy

Materials: NiTi/NiTiHf shape memory alloys, Ti-6Al-4V, functionally graded materials, metal powder characterization and reuse, superelastic and SME property tuning

Characterization: EBSD, in-situ XRD, DSC, WDS, SEM, DIC, FEA, thermomechanical and fatigue testing

Software: Python (data analysis, scientific visualization), MATLAB, CREO, AVEVA-PDMS

Standards & Qualification: ISO/ASTM AM standards, DoD/AFRL protocols, America Makes and LIFT consortium frameworks, aerospace and defense quality systems

AWARDS & FELLOWSHIPS

- 2nd Place, ASTM AM Student Poster Competition — ISO/ASTM TC261-F42 Joint Meeting, March 2025 (20+ entries)
- NSF INTERN Fellowship — Oak Ridge National Laboratory, Summer 2024
- Academic Distinction, Top 5% of class — M.Engg., Thapar Institute, 2020

SERVICE

- Judge, Student Research Week, Texas A&M University — evaluated 30 graduate and undergraduate research posters
- Reviewer and contributor, ISO/ASTM TC261-F42 additive manufacturing standards community